Chi-square tests and ANOVA CHEAT SHEET

ONE VARIANCE: CHI-SQUARE

$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$, df $n-1$ (population normal)

fund: n 24, s 4.8 %, σ_0 4 %: $\frac{23 \times 23.04}{16} = 33.12$

p = CHISQ.DIST.RT(33.12, 23) ≈ 0.079: do not reject

interval: $\frac{(n-1)s^2}{\chi^2(\text{upper})} \le \sigma^2 \le \frac{(n-1)s^2}{\chi^2(\text{lower})}$

38.076 and 11.689 (df 23): σ from 3.73 to 6.73 %

not symmetric: the interval sits off centre

lower tail (precise enough?): CHISQ.DIST(x, df, TRUE)

variance tests need normal data: plot first

TWO VARIANCES: F

$F = \frac{s_1^2}{s_2^2}$, df $n_1 - 1$ and $n_2 - 1$

lathes: $\frac{0.040^2}{0.025^2} = 2.56$ (15, 20), p ≈ 0.026

one tail: the variance H_a says is larger on top

two tails: larger s^2 on top, p = 2 × F.DIST.RT

F.TEST(range1, range2) returns the two-tailed p

THREE OR MORE PROPORTIONS

H_0: $p_1 = p_2 = \ldots = p_k$; $\bar{p}$ = pooled; $f_e = n \times \bar{p}$

$\chi^2 = \Sigma \frac{(f_o - f_e)^2}{f_e}$, df $k-1$

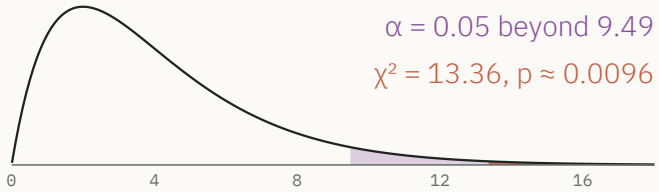
pages 12, 15, 16.2 %: $\chi^2 \approx 11.39$, p ≈ 0.0034

Marascuilo: $CV_{ij} = \sqrt{\chi^2_\alpha} \cdot \sqrt{(\bar{p}_i\bar{q}_i/n_i + \bar{p}_j\bar{q}_j/n_j)}$

1–3: 0.042 > 0.031 differ; 1–2: 0.030 < 0.0305

every expected count ≥ 5, or combine categories

TEST OF INDEPENDENCE



$f_e = \frac{\text{row total} \times \text{column total}}{n}$, df $(r-1)(c-1)$

channel × plan, 600 customers: $\chi^2 = 13.36$, df 4

p = CHISQ.TEST(observed, expected) ≈ 0.0096

read the cells: social → more basic, fewer premium

association, not cause

GOODNESS OF FIT

multinomial: $f_e = n \times$ stated share, df $k-1$

parcels 178, 124, 66, 32 vs 200, 120, 60, 20

$\chi^2 \approx 10.35$, p ≈ 0.016: the mix has shifted

normal: k equal-probability bins, $\bar{x} + z \cdot s$

df $k - 1 - 2$ (μ and σ estimated)

returns: $\chi^2 = 12.96$, df 5, p ≈ 0.024 (reject)

CHISQ.TEST would use df 7: p ≈ 0.073 (wrong)

WHICH DEGREES OF FREEDOM?

one variance: $n-1$ two variances: $n_1 - 1$, $n_2 - 1$

k proportions, multinomial fit: $k-1$

independence: $(r-1)(c-1)$

fit with m parameters estimated: $k - 1 - m$

READING A CHI-SQUARE

a big χ^2: the counts don't fit chance; not why

a tiny χ^2 (left tail): a fit too good to be true

Mendel 5,474 : 1,850 vs 3 : 1: $\chi^2 \approx 0.26$, p ≈ 0.61

Chi-square tests and ANOVA CHEAT SHEET

EXPERIMENTAL DESIGN

factor → treatments → experimental units → response

completely randomized: treatments assigned at random

experiment: we assign; observational: units choose

assume: normal, equal σ^2, independent observations

check: largest s $<$ 2 $\times$ smallest s

ONE-WAY ANOVA

$SSTR = \Sigma\, n_j(\bar{x}_j - \bar{\bar{x}})^2$, $MSTR = SSTR/(k-1)$

$SSE = \Sigma\,(n_j - 1)s_j^2$, $MSE = SSE/(n_T - k)$

$F = MSTR/MSE$; H_0 true → F near 1

$SST = SSTR + SSE$: $198 = 90 + 108$

layouts 41, 44, 47: $F = 45/9 = 5.0$, $p \approx 0.026$

twice the spread, same means: F 1.25, $p \approx 0.32$

one F test, not many t tests (3 at 0.05 → ≈ 0.14)

THE ANOVA TABLE

source	SS	df	MS	F	p
layouts	90	2	45	5.00	0.026
error	108	12	9		
total	198	14			

$p = F.DIST.RT(F, k-1, n_T - k)$

MULTIPLE COMPARISONS

$LSD = t_{\alpha/2}\sqrt{(MSE(1/n_i + 1/n_j))}$

layouts: $2.179 \times \sqrt{(9 \times 2/5)} \approx 4.13$ → only 1–3 (6)

interval: $\bar{x}_i - \bar{x}_j \pm LSD$: $6 \pm 4.13 = 1.87 \ldots 10.13$

experimentwise: $1 - (1-\alpha)^C$; C = 3 → 0.143, 10 → 0.401

Bonferroni: α/C each; 3 pairs: 0.0167, LSD 5.27

RANDOMIZED BLOCKS

$SST = SSTR + SSBL + SSE$; df error $(k-1)(b-1)$

6 couriers $\times$ 3 planners: SSBL 600 out of the error

blocked: $F = 18/2.4 = 7.5$, $p \approx 0.010$

ignoring blocks: $F \approx 0.43$, $p \approx 0.66$

FACTORIAL EXPERIMENTS

$SST = SSA + SSB + SSAB + SSE$; df error $ab(r-1)$

discount $\times$ email: cell means 20, 22, 23, 31

interaction $F = 27/2.75 \approx 9.82$, $p \approx 0.014$

lines not parallel: read effects through it

CASE: TARNWICK (FICTIONAL)

4 suppliers $\times$ 24 batches: strength, rejects / 500

Cedar: highest mean 69.0, but s 5.0 $>$ 3: $\chi^2 \approx 63.8$

rejects 2.13, 2.13, 3.33, 2.29 %: $\chi^2 \approx 49.9$

3 others: $F \approx 12.8$; Alder–Birch $p \approx 0.028 > 0.0167$

choose Birch: $60,000 a year cheaper than Alder

SPREADSHEET AND PYTHON

CHISQ.DIST(x, df, TRUE) · CHISQ.DIST.RT(x, df)

CHISQ.INV(p, df) · CHISQ.INV.RT(p, df)

CHISQ.TEST(observed, expected)

F.DIST.RT(F, df1, df2) · F.INV.RT(α, df1, df2)

F.TEST(r1, r2) (two tails) · DEVSQ · VAR.S

T.INV.2T(α, df) for the LSD

Python (scipy.stats): chi2, f, chi2_contingency, chisquare, f_oneway